

PSY 308/508

Cognition of Motor Movement

A very warm welcome to you all !!!!!!!!!!!

Team

- **Instructor: Sonia Baloni Ray**
- **TA:**
 - **Shruti Khurana**
 - **Sangramjit Maity**

Office hours:

Sonia Baloni Ray: Fridays 12:45pm- 1:45pm

TA: — — — — — — — To be announced

The course

Lectures

- **Tuesday & Friday@ 11:00am- 12:30pm (A007)**
- **Live lectures**
 - **Scanned chapters and lecture slides will be provided immediately after the classes.**
- **Google Classroom Code: provide next week**

Assessment Plan (300)

- 2 Assignments (15%)
 - **Paper Writing** (3rd Week): 28th August
 - **Data Analysis: Project** (10th Week): 13th October
- 3 Quizzes (15%)
 - **Quiz 1**: 5th Week: 8th September
 - **Quiz 2**: 13th Week: 23rd October
 - **Quiz 3**: 15th Week : 13th November
- **Mid-Semester Exam** (35%)
- **End-Semester Exam** (35%)

Assessment Plan (500)

- **Project (25%):** Replicate a study provided to you (2nd Week):
 - **Project Report + presentation** —> *To be Announced*
- **2 Assignments (15%)**
 - **Paper Writing** (3rd Week): *28th August*
 - **Commentary** on a paper (10th Week) : *13th October*
- **3 Quizzes (15%)**
 - Quiz 1: 5th Week: *8th September*
 - Quiz 2: 13th Week: *23rd October*
 - Quiz 3: 15th Week : *13th November*
- **Mid-Semester Exam (25%)**
- **End-Semester Exam (25%)**

Project: 5XX

- **Full support in data collection and design of experiment**
- **Eye tracker / behavioural experiments**
- **Timely submission**
- **Last moment dependencies**
- **Marking**
 - **Rubric will be given**
 - **Nature of analysis**
 - **Quality of data representation**

Project: 3XX

- **Curated data will be given appropriately formatted.**
- **Eye tracker / behavioural experiments**
- **Timely submission**
- **Last moment dependencies**
- **Marking**
 - **Rubric will be given**
 - **Nature of analysis**
 - **Quality of data representation**

Mid and End Semester Examinations

- **70 % : Scoring**
 - MCQs
 - Fill In the blanks
 - True/False
 - Short Answer questions
 - Definitions
 - Short questions
 - Differences/similarities
- **30%: Long Answer questions**

Class material

- Google classroom
- Share the slides (only for reference)
- Books and Research papers
 - Only soft copy
- Attendance will be taken, **NOT** marked

Evaluations

- Google classroom update with grades
 - Please **regularly** check if the marks are updated.
- Copying contents of slides and expect 100%
- Deadline extension
- Attendance

Missing Quiz/ Mid-Sem/End-Sem

- **Assessment will be taken on the last class (For quizzes)**
- **Long answer questions only (For Mid and End semester exams)**

Final Grading

- **Relative**
- **mean $\pm$ standard deviation**
- **No manual cut-off**

Grade	Range
A	marks $\geq$ (meanPer + (1.5*stdPer))
A-	(meanPer + stdPer) - (meanPer + (1.5*stdPer))
B	(meanPer + (0.5*stdPer)) - (meanPer + stdPer)
B-	meanPer - (meanPer + (0.5*stdPer))
C	(meanPer - (0.5*stdPer)) - meanPer
C-	(meanPer - stdPer) - (meanPer - (0.5*stdPer))
D	(meanPer - (2.5*stdPer)) - (meanPer - stdPer)
F	marks $<$ (meanPer - (2.5*stdPer))

Plagiarism

- Website inspired —> language change: Not accepted
- Bound to report plag to DoAA: No exception
- Always submit assignment after checking through library approved s/w: turnitin, Orkund.
- less than 10% acceptable
- -1 for each 5% additional plag
- $\geq 20\%$ == 0 and bound to report to DoAA.
- No exceptions

<https://www.iiitd.ac.in/academics/resources/acad-dishonesty-faqs>

Motor Movement



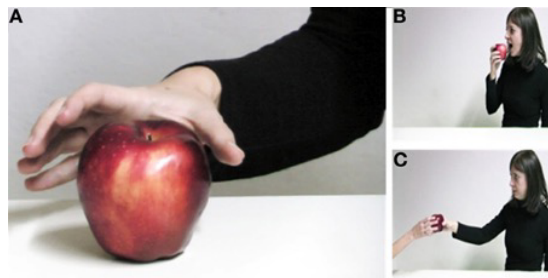
<https://www.biologyonline.com/tutorials/control-of-body-movement>



<https://www.vccafe.com/2017/07/31/creating-a-movement/>



<http://nupurnritya.com/courses/degree-courses-in-kathak/>



Becchio et al, 2012



<https://www.morganmckinley.com.au/article/all-important-first-handshake>



<https://www.thestatesman.com/entertainment/farhan-akhtar-croons-telugu-song-for-mahesh-babus-bharat-ane-nenu-1502615016.html>

<https://www.musicradar.com/how-to/improve-your-acoustic-folk-guitar-playing-in-20-minutes>

Why study motor cognition?



<https://blogs.3ds.com/northamerica/future-robots-and-ensuring-human-safety/>

NASA: Valkyrie



<https://skyandtelescope.org/astronomy-news/meet-valkyrie-nasa-space-robot/>

F.E.D.O.R: Skybot



<https://www.space.com/russia-launching-humanoid-robot-into-space.html>

Sophia



<https://www.independent.co.uk/life-style/gadgets-and-tech/sophia-robot-launch-hanson-b1792123.html>

Cognition

<https://www.youtube.com/watch?v=fn3KWM1kuAw>

<https://www.youtube.com/watch?v=yLtdzJ6mVMk&t=146s> →

Alan Turing lecture

Prosthetics



<https://www.sralab.org/center-bionic-medicine/powered-prosthetic-arm>



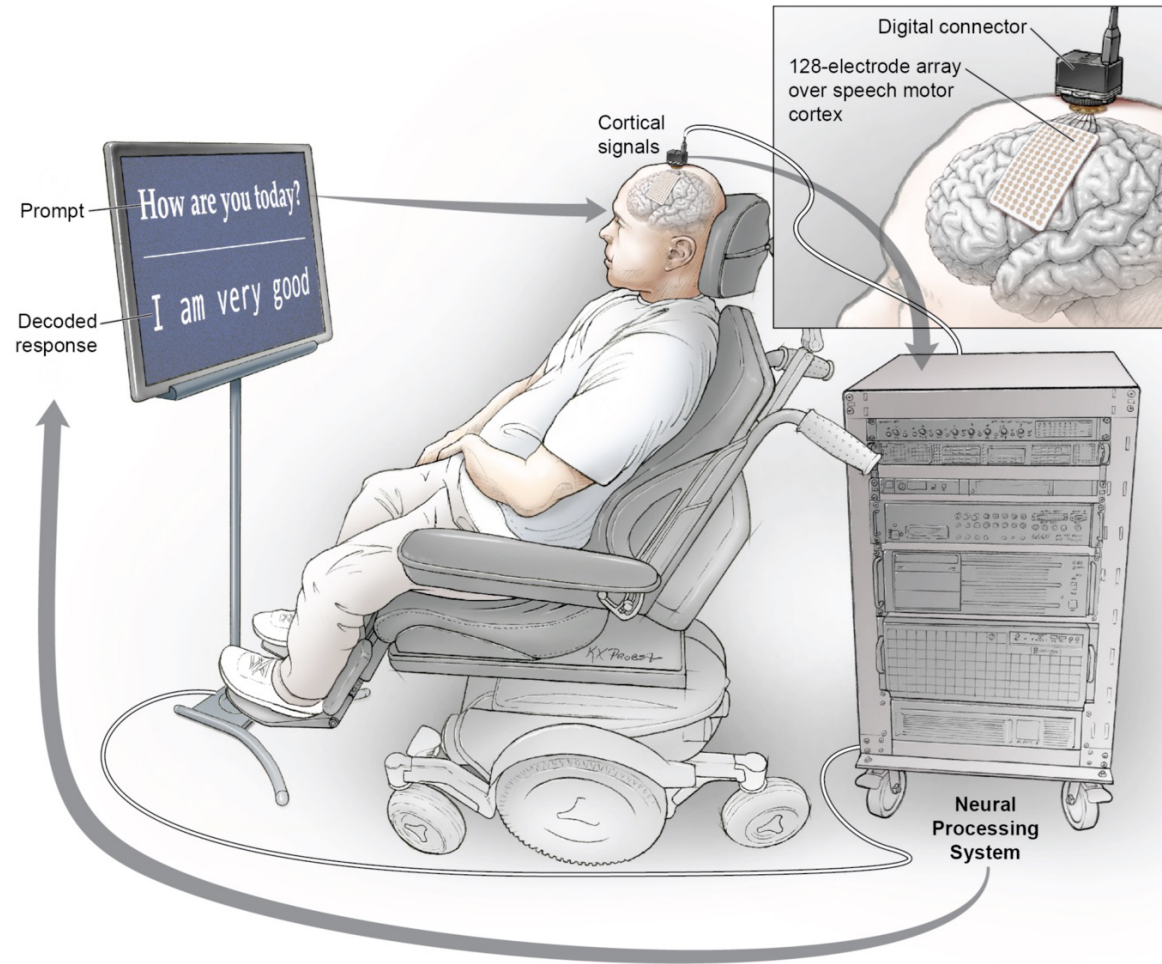
<https://www.engadget.com/2019-09-09-eth-zurich-prosthetic-leg-sense-of-touch.html>



https://www.thestar.com/sports/2013/02/14/blade_runner_oscar_pistorius_charged_with_murder_in_shooting_of_girlfriend.html

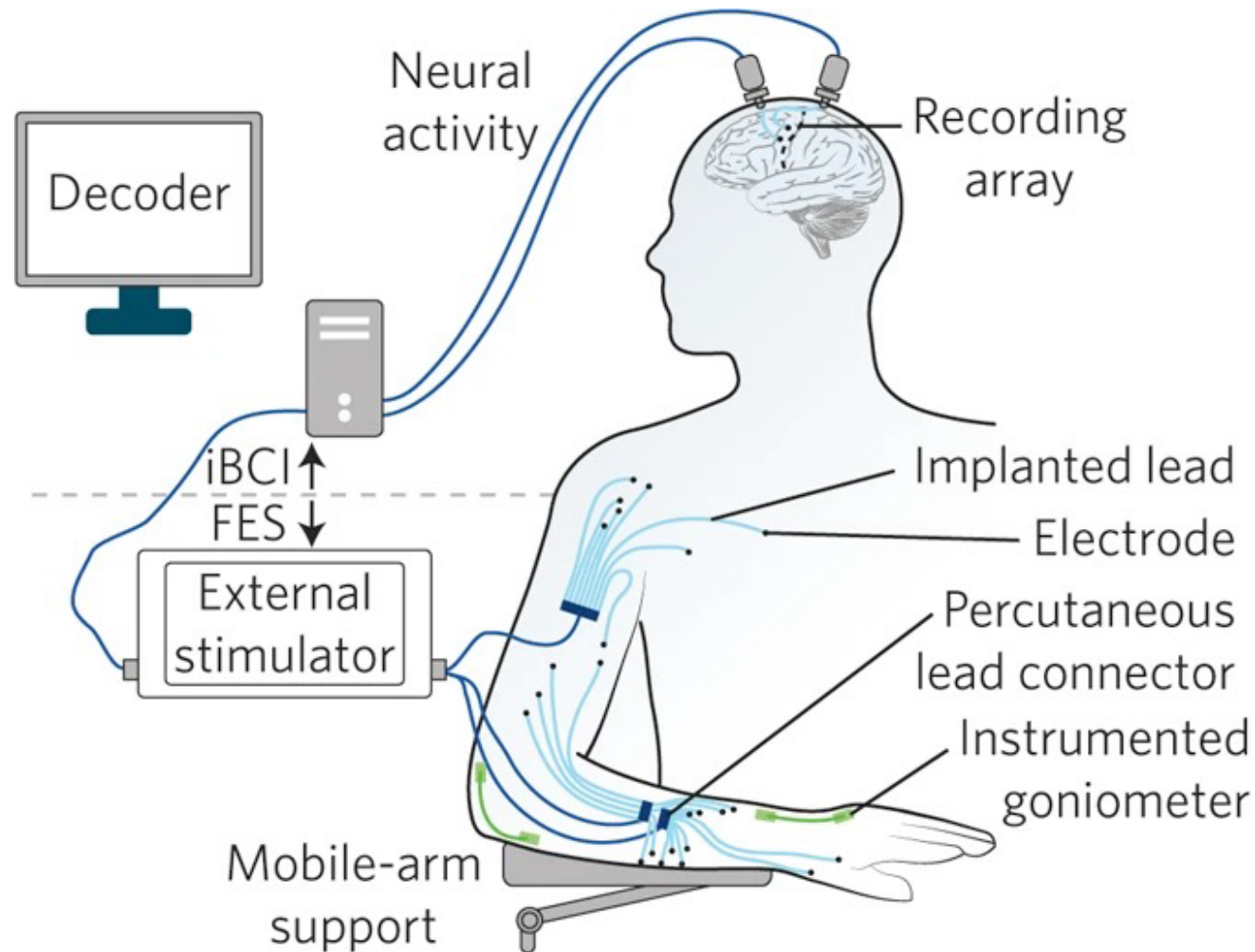
Neuroprosthetics

Edward Chang, UCSF



<https://changlab.ucsf.edu/publications/speech-lab>

Limb Neuroprosthetics



Why study motor cognition?

Robots



<https://blogs.3ds.com/northamerica/future-robots-and-ensuring-human-safety/>

Social Robots



<https://www.independent.co.uk/life-style/gadgets-and-tech/sophia-robot-launch-hanson-b1792123.html>

Neuroprosthetics



<https://www.sralab.org/center-bionic-medicine/powered-prosthetic-arm>

Topics

- Introduction to brain
- Methodologies
 - Experimentation
 - Data Collection
- Types of action
 - Spatial Navigation
 - Eye movement
 - Hand movement
 - Eye-hand coordination
 - Reaching/Grasping
 - Drawing/writing/keyboarding
 - Speech